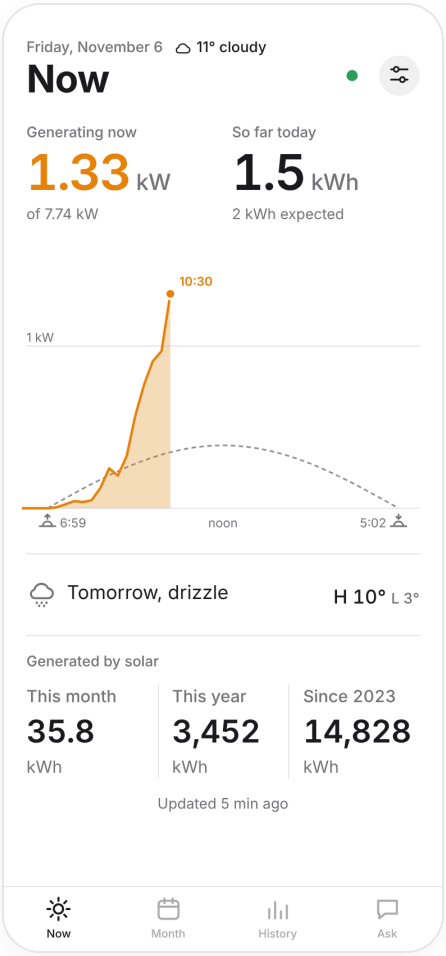
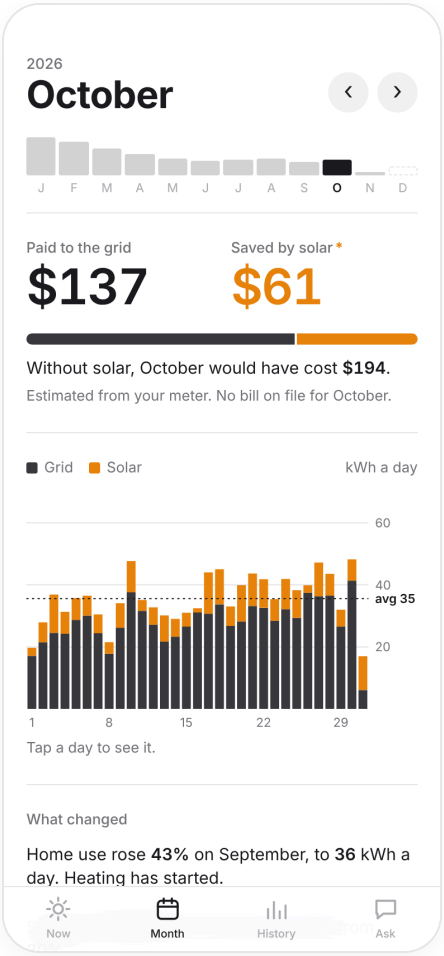
Solar

A private app for one home: what the panels make, what power costs, and what solar saves.

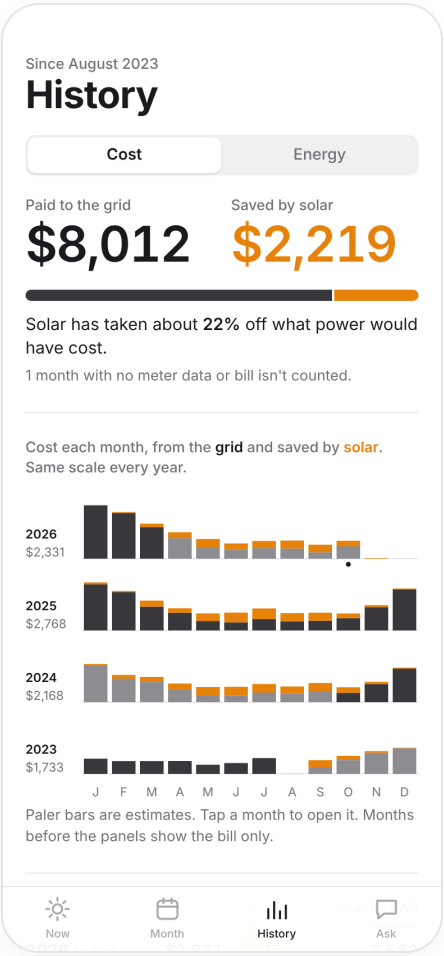
Solar joins two sources that never talk to each other: the **SolarEdge inverter** on the house (17 panels, 7.7 kW) and **Toronto Hydro's** meter readings and bills. It works out Ontario's net metering rules the way the bills do, so the savings are real dollars, checked against every bill on file. It opens to four tabs, Now, Month, History and **Ask**, where you can ask about your energy by typing or by voice, and it runs from the iPhone Home Screen.



Now
What the panels are making, today so far against the expected curve, sunrise and sunset, weather and the totals since 2023. A dot shows the system's health.



Month
What you paid and what solar saved, with what the month would have cost without it. Every day split into grid and solar. Tap a day or open the bill.



History
Every month since January 2023, one row per year on one scale. Months before the panels show the bill only. Switch to Energy for what the panels made.

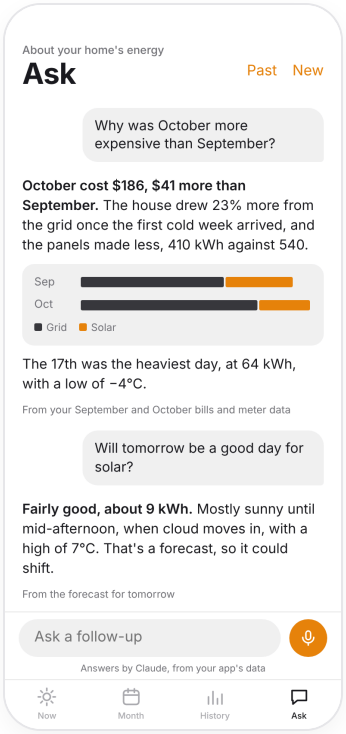
AT A GLANCE (SAMPLE FIGURES)

14,828
kWh generated since August 2023

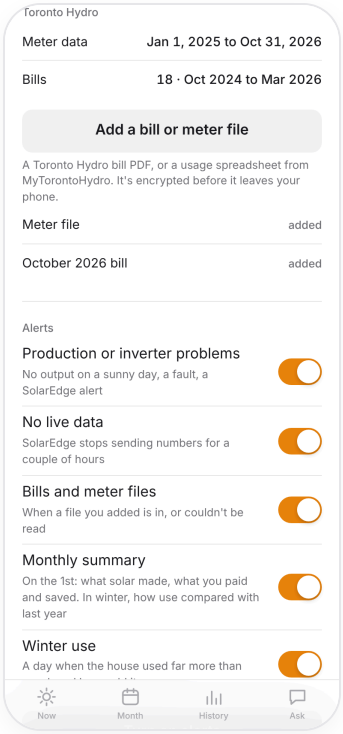
$2,219
saved by solar, about 22% off the power bill

Every
Toronto Hydro bill checked line by line against the meter

1 hr
between updates, with alerts when something's wrong



Ask
Questions in plain words, answered from the app's own data, with a small chart when it helps.



Settings
Add a bill or meter file from the phone, and choose which alerts to get.

Ask

Type a question or tap the mic and say it. Claude answers in a few plain sentences with the key number first, from your months, bills, the last 120 days and the live forecast, and names the data it used. Spoken questions are answered aloud. Conversations are kept on the phone under Past.

Live from the inverter

Power now, today's output and the expected curve, which levels off at the inverter's 6 kW limit. At night a small moon shows the real phase; tap it for its name.

Costs that match the bills

Credits for power sent back are valued at the time-of-use price when it went out, delivery is charged on net use, and HST and the Ontario rebate are applied as Toronto Hydro does. Months without a bill are estimated from the meter, usually within a few percent.

Bills and meter data

Add a bill PDF or a MyTorontoHydro spreadsheet from Settings. It's encrypted on the phone, read by the next update, and a bill is only added if its lines add up to the printed total. The PDF itself is never kept.

Alerts

Phone notifications when production stops on a sunny day or the inverter reports a fault, when SolarEdge stops sending data, when a file is added, a summary on the 1st of each month, and in winter, days of unusually heavy use with the temperature.

Private by design

All data is encrypted with a passphrase before it's published. Unlock with the passphrase, or with Face ID on a trusted phone. Serial numbers and tokens live only inside the encrypted data.

How it runs

A scheduled job on GitHub fetches SolarEdge and the weather about every hour, reads any new files, rebuilds the site on GitHub Pages and sends alerts. There's no server to maintain.

Your system

Panels, optimizers, inverter, rapid shutdown switch, meter and installer, with serial numbers for warranty or service, and what to do in an emergency.

Good to know

Meter data comes from exported files, so costs trail by however long it's been since the last one. SolarEdge's API has no per-panel data. An answer in Ask is lost if you leave Solar mid-answer; it's asked again when you come back.

SOLAR

Technical setup

SOURCES AND LIBRARIES

PIECE	VERSION	WHAT IT DOES	LOADED FROM	LICENCE
App code	App 1.53	The four tabs, charts, net metering maths, encryption and Ask (`app.js` , `lock.js` , `style.css` , `sw.js`). No framework; charts are hand-drawn SVG	GitHub Pages, this repo's `web/`	Written for the app
Build job	Python 3	Fetches data, reads bills and meter files, models costs, encrypts, sends alerts (`scripts/build.py` , `money.py` , `parse_bill.py`)	GitHub Actions	Written for the app
Transformers.js	3.5.1	Runs the speech model in the browser, with ONNX Runtime Web inside	cdn.jsdelivr.net	Apache-2.0
Whisper tiny.en	Xenova, 8-bit	Turns a spoken question into text on the phone (about 40 MB, cached after first use)	huggingface.co	MIT
Python libraries	latest	cryptography (AES-GCM), openpyxl (meter spreadsheets), pywebpush (notifications)	PyPI	Apache-2.0, MIT, MPL-2.0
Poppler pdftotext	Ubuntu package	Reads the text of bill PDFs	apt	GPL-2.0
Type and icons	n/a	System font; icons drawn as SVG for the app	The device	Written for the app

INTEGRATIONS

SERVICE	USED FOR	DETAILS
SolarEdge Monitoring API v2	Production, power now, inverter status and alerts	`monitoringapi.solaredge.com/v2` , OAuth through `connect.solaredge.com` ; the refresh token rotates and is saved back as a GitHub secret. Free tier, 2,000 calls a month
Toronto Hydro	Bills and hourly meter readings	No API. Bill PDFs and MyTorontoHydro spreadsheets are added from Settings and read by the build; the PDF isn't kept
Open-Meteo	Weather now, the forecast, and past weather	`api.open-meteo.com` and `archive-api.open-meteo.com` , no key. Called by the build, the Now screen, and Ask for each question
Anthropic Claude API	Answers in Ask	`api.anthropic.com/v1/messages` , model `claude-haiku-5-5` , called from the phone. Sends the question, recent conversation and a summary of months, bills, 120 days and the forecast. Key in the `ANTHROPIC_API_KEY` secret, carried inside the encrypted data
GitHub	Hosting, the hourly job, uploads	Pages serves the site; Actions runs the build hourly; files from the phone are committed encrypted to `data/inbox/` through the Git Data API with a fine-grained token (`UPLOAD_TOKEN`)
Web Push (Apple)	Alerts on the phone	VAPID-signed pushes from the build; subscriptions kept encrypted in `data/push.enc`
Browser features	Unlock, voice, offline	Web Crypto (PBKDF2-SHA256, AES-GCM), passkeys for Face ID unlock, microphone and Web Audio for recording, speech synthesis to read answers, a service worker, and local storage for Ask conversations

What runs where

On the phone: unlocking and decrypting the data, every screen and chart, recording and transcribing questions, reading answers aloud, and the calls to Claude and Open-Meteo. Nothing else leaves the phone except files you add, which are encrypted first.

On GitHub, about every hour: the build fetches SolarEdge and the weather, reads new files, rebuilds the site and sends alerts. The site's data is published encrypted with a key from the passphrase (PBKDF2-SHA256, 600,000 rounds); serial numbers and tokens exist only inside it.

This document: `docs/overview/` . Make sample data with `fake_data.py` , take screenshots with `shoot.js` , print with `pdf.js` .